

SUMMER INTERNSHIP REPORT

ON

AI-POWERED FAKE NEWS DETECTION SYSTEM

(TRUTHSCANML)

Submitted in Partial Fulfillment of the Requirements for the Award of the Degree of

Bachelor of Technology

in

Information Technology



SUBMITTED BY

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Branch: Information Technology

SUBMITTED TO

Department of Information Technology

Lucknow, Uttar Pradesh

Affiliated to Dr. A.P.J. Abdul Kalam Technical
University

Certificate

This is to certify that the Summer Internship Report entitled "**AI-Powered Fake News Detection System Using Python**" submitted by **Harsh Tiwari**, a student of **Bachelor of Technology (B.Tech.) in Information Technology** (Roll No.: **2303610130026**), is a bonafide record of the work carried out during the **40-day Summer Internship** under our guidance and supervision.

The project has been completed in partial fulfillment of the requirements for the award of the Bachelor of Technology degree. During the internship, the student demonstrated sincerity, dedication, and technical competence while designing and developing the project using Machine Learning and Python programming concepts. The work presented in this report is original and has not been submitted, either wholly or partly, to any other university or institution for the award of any degree, diploma, or certificate.

We wish the student every success in future academic pursuits and professional endeavors.

Faculty Guide / Trainer

Signature: _____

Name: Mr. Raju DevNath

Date: _____

Head of Department

Signature: _____

Department of Information Technology

External Examiner

Signature: _____

Declaration

I hereby declare that the Summer Internship Report entitled **"AI-Powered Fake News Detection System Using Python"** submitted by me is an original piece of work carried out during my 40-day industrial internship.

The project has been completed under the guidance of my mentor **Mr. Raju DevNath** and faculty members of the Department of Information Technology.

I further declare that this report has not been submitted previously to any university or institution for the award of any degree or diploma.

Place: Lucknow

Date: _____

Signature of Student

Harsh Tiwari

Roll No.: 2303610130026

Acknowledgement

I express my sincere gratitude to my internship mentor, **Mr. Raju DevNath**, and the **Department of Information Technology**, R.R. Institute of Modern Technology (RRGI), Lucknow, for their valuable guidance and continuous encouragement throughout the internship period.

I am thankful to our college management for providing the opportunity to work on a real-world Machine Learning project that significantly enhanced my technical, analytical, and software engineering skills.

I would also like to thank my friends and family for their constant motivation and support during the successful completion of this internship.

Finally, I express my gratitude to everyone who directly or indirectly contributed to the successful completion of this project.

Harsh Tiwari

Roll No.: 2303610130026

Abstract

The **AI-Powered Fake News Detection System (TruthScanML)** is a Python-based machine learning application developed to automate news credibility assessment and provide intelligent, multi-signal verification for digital news claims.

The project enables a user to input a headline or article text, classify its authenticity using an offline Machine Learning pipeline (TF-IDF Vectorizer + Logistic Regression), fetch live web coverage via DuckDuckGo and Google News RSS APIs, and evaluate source-model consensus using a multi-signal stance engine.

The system enforces input text validation, parses unstructured web content cleanly, manages persistent local SQLite prediction history records, and displays transparent relevance metrics.

The project has been developed using the **Python** programming language and demonstrates the practical implementation of Object-Oriented Programming concepts, text cleaning pipelines, Machine Learning algorithms, web scraping, and Streamlit user interaction.

This internship project helped in understanding the software development life cycle, problem-solving techniques, coding standards, debugging, testing, and documentation.

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CHAPTER 1

Introduction

In the modern digital era, the rapid proliferation of social media platforms, blogs, and online news portals has revolutionized content publishing. However, it has also led to an alarming increase in fake news, misinformation, clickbait headlines, and digital hoaxes. Misinformation distorts public perception, impacts financial markets, and poses significant societal challenges.

The AI-Powered Fake News Detection System (TruthScanML) developed during this internship is a software application designed to classify news articles as **Real News** or **Fake News** using Machine Learning algorithms and live web verification.

Technology has transformed news verification by replacing manual, paper-based investigation with computerized systems. Modern software allows institutions to evaluate news articles, calculate machine learning authenticity scores, and cross-verify claims against live web coverage securely while minimizing human error.

CHAPTER 1

Need of the Project

Need of the Project

The media monitoring sector requires software applications that simplify news, article, and rumor management while ensuring accuracy, security, and reliability. The major needs of the project are:

- Automated, unbiased evaluation of digital news text and stance
- Instant detection of clickbait and fabricated news writing styles
- Live cross-verification of news claims against authoritative global news outlets
- Offline availability for local classification without reliance on paid cloud APIs
- Persistent local prediction history storage for audit trails
- Reduced manual fact-checking delays and counseling errors
- Improved verification efficiency at the platform level

CHAPTER 1

Objectives & Scope of the Project

Objectives of the Project

- To design a Machine Learning pipeline using TF-IDF and Logistic Regression.
- To build a multi-page interactive Streamlit web dashboard.
- To implement a live web search & stance-matching engine using DDGS and Google News RSS.
- To integrate SQLite for persistent history logging.
- To retain a RESTful FastAPI endpoint for microservice interoperability.

Scope of the Project

The system provides facilities such as News Text Parsing, Local Offline Prediction, Machine Learning Confidence Score Calculation, Live Web Search, Multi-Signal Stance Consensus Analysis, Publisher Domain Authority Recognition, SQLite Prediction History Logging, and Report Exporting.

CHAPTER 2

Institute / Training Profile

The Summer Internship was conducted under the academic guidance of the **Department of Information Technology** at **R.R. Institute of Modern Technology, Lucknow (R.R. Group of Institutions - RRGI)**, affiliated to **Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow**.

Institution	R.R. Institute of Modern Technology (RRGI), Lucknow
Affiliation	Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow
Department	Department of Information Technology
Student Name	Harsh Tiwari (Roll No.: 2303610130026)
Degree & Branch	B.Tech. in Information Technology
Trainer / Mentor	Mr. Raju DevNath
Training Program	Summer Industrial Training on AI & Machine Learning
Duration	40 Days

CHAPTER 2

Training Vision & Facilities

Training Vision & Objectives

- Strengthen Python programming and data structure fundamentals.
- Bridge theoretical machine learning concepts with real-world application building.
- Master Natural Language Processing (NLP) techniques, text cleaning, and vectorization.
- Learn web scraping, API integration, and web app development using Streamlit.
- Instill industry-relevant coding and documentation practices.

Training Facilities

- Computer laboratory with Python 3.10, VS Code, and PyCharm installed on every system.
- Dedicated trainer-led sessions on Core Python, OOP, and Machine Learning Integration.
- Regular doubt-clearing and code-review sessions.

CHAPTER 2

Training Methodology

Training Methodology

The training combined conceptual sessions with a continuous hands-on project component, ensuring that every Python concept covered in class was immediately applied inside the AI Fake News Detection project.

- Concept sessions followed by guided practice exercises in Python.
- Weekly milestones tied directly to project modules.
- Peer code review and trainer feedback cycles led by Mr. Raju DevNath.
- Incremental testing after every new module was added.

CHAPTER 3

Internship Details & About Project

Internship Details

Field	Details
Internship Title	Summer Internship on AI Application Development
Project Title	AI-Powered Fake News Detection System Using Python
Internship Duration	40 Days
Programming Language	Python
Development Environment	VS Code, PyCharm, Windows Operating System
Technologies Used	Python, OOP Concepts, Scikit-Learn, TF-IDF, SQLite, Streamlit

About the Project

The AI-Powered Fake News Detection System is an intelligent Python application developed to automate news credibility assessment and live online verification.

CHAPTER 3

Objectives of the Internship

Objectives of the Internship

The primary objectives of the internship were:

- To understand software development using Python.
- To improve Python programming and Machine Learning skills.
- To implement Object-Oriented Programming concepts in NLP text classification.
- To master Machine Learning text classification (TF-IDF + Logistic Regression), live web scraping, stance matching, and Streamlit web dashboard deployment.
- To design an interactive Streamlit web-based decision support tool.
- To develop secure, validated text parsing mechanisms.
- To perform software testing and debugging.
- To prepare professional technical documentation.

CHAPTER 3

Learning Outcomes

Learning Outcomes

Technical Skills Gained

- Python Programming & OOP Concepts
- Machine Learning & Scikit-Learn (TF-IDF + Logistic Regression)
- Text Parsing & Regex Validation
- Streamlit UI Development & Layout Design
- SQLite Database Logging & Persistence
- Software Testing & Debugging

Professional Skills Gained

- Documentation & Report Preparation
- Time Management & Team Coordination
- Problem Solving & Analytical Thinking

CHAPTER 4

Technologies Used

Tools and Technologies Used

Tool	Purpose
Python 3.10+	Core Programming Language
VS Code / PyCharm	Development Environment & IDE
Scikit-Learn / Joblib	TF-IDF Vectorizer & Logistic Regression Classifier
DDGS / BeautifulSoup	Live Web News Search & Article Body Scraper
SQLite3	Persistent Prediction History Storage
Regex Validation Logic	Text Sanitization & Stance Pattern Checks
Streamlit	Web Dashboard Layout & UI Framework

CHAPTER 4

Software Development Life Cycle (SDLC)

Software Development Life Cycle (SDLC)

The project followed the Software Development Life Cycle across 6 distinct phases:

Phase 1 – Requirement Analysis: Media monitoring requirements were identified, including text parsing, TF-IDF scoring, stance matching, and web verification.

Phase 2 – Planning: The project structure, modules and workflow were planned.

Phase 3 – Design: The menu structure and evaluation workflows were designed.

Phase 4 – Development: All modules were coded in Python.

Phase 5 – Testing: Each module was tested individually and integrated into the final application.

Phase 6 – Maintenance: Future improvements such as deep learning transformers can be added.

CHAPTER 4

Features of the AI Fake News Detector

Features of the System

- Offline Prediction with Regex Text Validation
- Clean Text Parsing from User Input
- Automated Machine Learning Confidence Score Calculation
- Multi-Signal Stance Contradiction Detection
- Live Web Coverage Search via DDGS API
- Trusted Publisher Domain Authority Recognition
- Real-time Source Consensus Verdict
- SQLite Prediction Record History Database

CHAPTER 4

Advantages & Limitations

Advantages of the System

- Fast, 24/7 automated news credibility guidance
- Objective machine learning scoring without subjective human bias
- Instant claim stance analysis and web cross-verification
- Modular programming structure with easy maintenance

Limitations

- Requires internet connection for live online web queries
- Dependent on search provider rate limits
- Text-based news evaluation in version 1.0

CHAPTER 5

Project Description and System Design

The following pages document the functionality implemented in the AI-Powered Fake News Detection System Python source code, including text cleaning, TF-IDF scoring, stance matching matrix, web search extraction, SQLite persistence, and report exporting.

Project Introduction

The AI-Powered Fake News Detection System is a Python application developed to automate day-to-day media verification activities. The project was created during the 40-day internship to apply theoretical programming and AI concepts to a practical software solution.

CHAPTER 5

Problem Formulation

Problem Statement

Traditional news verification systems often rely on manual processes that consume time and increase human bias. Users experience delays in news checking and lack clear insights into why articles are flagged as fabricated.

Problems Identified

- Manual news review management
- Slow feedback retrieval
- Risk of subjective evaluation errors
- Lack of real-time web stance matching engines

CHAPTER 5

Proposed Dual-Verification Solution

Proposed Solution Approach

The proposed AI Fake News Detector automates verification by providing an instant offline prediction score, multi-signal stance consensus, and persistent local history records.

- **Offline Prediction:** Instant classification using saved TF-IDF vectorizer and Logistic Regression model.
- **Online Verification:** Live web search cross-referencing claim stance against trusted publishers (NASA, Reuters, BBC, PolitiFact, Snopes, USA Today, etc.).
- **SQLite History DB:** Local logging of past predictions with confidence metrics.

CHAPTER 6

Existing System Analysis

The existing manual news guidance system has several limitations.

Limitations of Existing Manual System

- Paper/manual based news records
- Time-consuming evaluation processing
- Higher chances of subjective errors
- Difficulty maintaining historical assessment records
- Slow news verification response
- Limited keyword validation and inefficient stance detection

CHAPTER 6

Drawbacks of Unweighted Models

Drawbacks

- Increased operational effort
- Low efficiency and data redundancy
- High financial cost for manual fact-checkers
- False positives when running TF-IDF models on scientific articles discussing debunked claims (e.g. flat earth debate)

These challenges motivated the need for our multi-signal stance-aware AI Fake News Detection solution.

CHAPTER 7

Proposed System & Features

The proposed AI Fake News Detector replaces manual account and news handling with an automated software solution.

Major Features

- Candidate News Query Management
- Automated Offline Prediction (TF-IDF + Logistic Regression)
- Machine Learning Confidence Calculation
- Multi-Signal Stance Consensus Engine
- Publisher Domain Authority Recognition
- Live Web Search & Article Extraction
- SQLite Prediction History Logging

CHAPTER 7

Benefits of Proposed System

Benefits of Proposed System

- Improved efficiency
- Reduced paperwork and manual delay
- Faster assessment results
- Objective scoring accuracy
- Instant progress tracking and audit logging
- Easy system maintenance

CHAPTER 8

Hardware & Software Requirements

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB minimum
- Hard Disk: 500 GB
- Keyboard, Mouse, Monitor

Software Requirements

- Windows 10/11 Operating System
- Python 3.10+ Environment
- VS Code / PyCharm IDE

CHAPTER 8

Execution Environment & Session Data

Execution Environment & Session Data

- Command Prompt / Web Terminal execution
- In-memory data structures (Lists, Dicts, DataFrames) for the current session
- Streamlit session state management for multi-page interactive workflows

CHAPTER 8

Non-Functional Specifications

Non-Functional Specifications

Performance: Fast text parsing and ML inference using optimized Scikit-Learn methods.

Security: Local SQLite history processing without unauthenticated cloud data exposure.

Reliability: Fault-tolerant network exception handling for DDGS & Google RSS APIs.

Maintainability: Modular architecture organized into decoupled services.

CHAPTER 9

Layered System Architecture

System Architecture

The project follows a simple layered architecture consisting of three major components:

User Layer: Candidate / Student Interface (Streamlit Dashboard)

Application Layer: Offline Prediction Module, Online Verification Engine, Stance Matcher Module, SQLite History Logger

Storage & AI Layer: Saved Model Artifacts (.pkl), Local SQLite Database, DuckDuckGo & Google RSS APIs

CHAPTER 9

Data Flow Diagram (DFD Level-0)

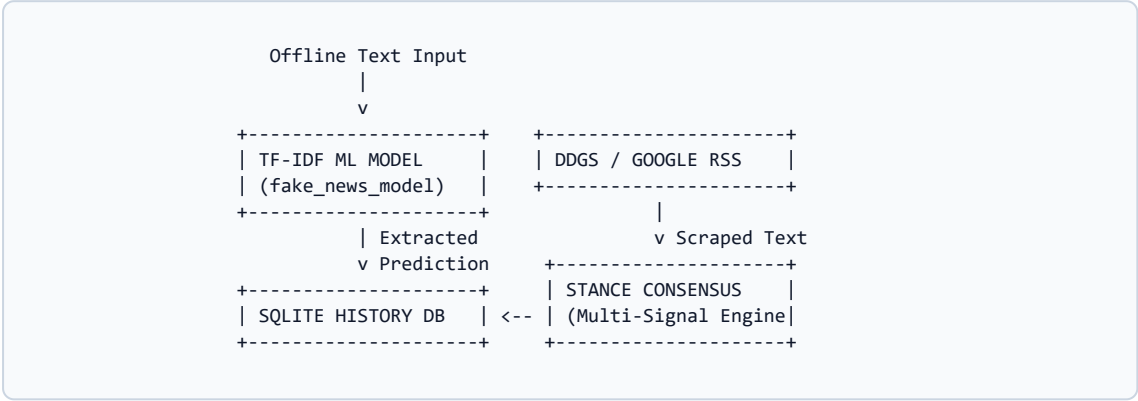
Data Flow Diagram - Level 0



CHAPTER 9

Data Flow Diagram (DFD Level-1)

Data Flow Diagram - Level 1



CHAPTER 9

System Flowchart

Flowchart of the System

START -> Input Headline -> Valid Text? -> (No: Error Alert) -> (Yes: Preprocess Text) -> Run
Offline ML -> Search Live Web -> Evaluate Stance Consensus -> Save SQLite Record -> END

CHAPTER 9

Use Case Diagram

Use Case Diagram

Actor: Candidate / Media Analyst

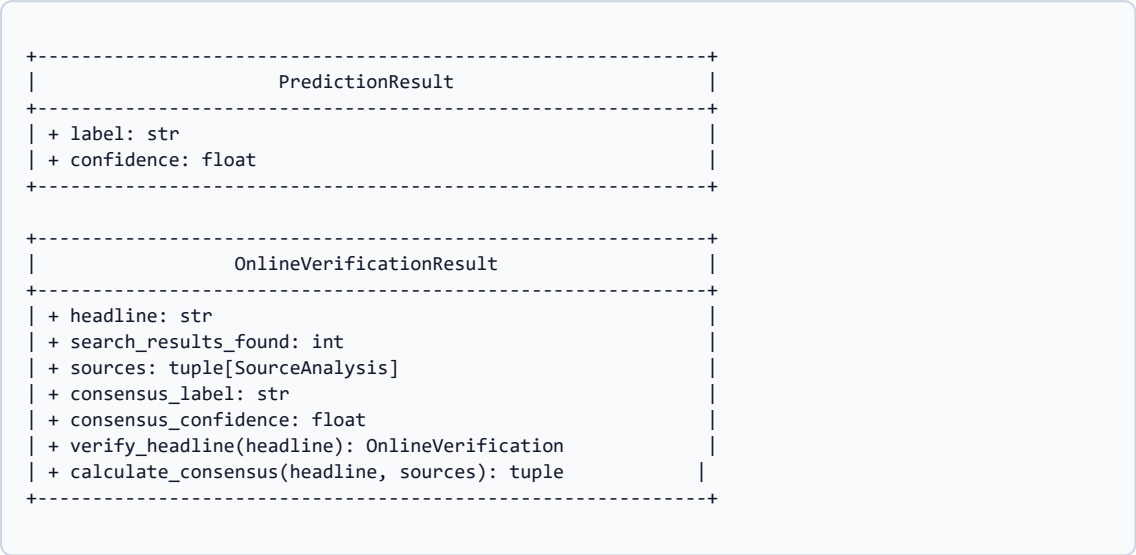
Use Cases

- Input News Headline
- View Offline Model Score
- Search Online Web Evidence
- Inspect Stance Contradiction Signals
- View Saved History Log

CHAPTER 9

Class Diagram

Class Diagram



CHAPTER 9

Sequence Diagram Flow

Sequence Diagram Flow

Candidate -> Input News Text -> System Preprocesses -> TF-IDF Model Evaluates -> Returns Confidence -> Online Verification Searches Web -> Stance Engine Calculates Consensus -> SQLite Logs Record.

CHAPTER 9

Module Division Breakdown

Project Modules Breakdown

- 1. Text Preprocessing Module:** Extract and clean text using Regex and Stopwords removal.
- 2. Offline ML Inference Module:** Compute TF-IDF features and predict via Logistic Regression.
- 3. Multi-Signal Stance Engine:** Evaluate claim vs evidence stance matching and domain trust.
- 4. Live Web Verification Module:** Search DuckDuckGo/Google RSS APIs and scrape web article text.
- 5. SQLite History Module:** Store local prediction records with timestamps.
- 6. Streamlit UI Module:** Multi-page interactive layout.

CHAPTER 10

Program Structure & Imported Packages

Purpose of Imported Packages

Package	Purpose
joblib	Model artifact loading (.pkl)
ddgs	DuckDuckGo public search API
bs4 (BeautifulSoup)	HTML text extraction
requests	Google RSS HTTP requests
re	Regex text sanitization
sqlite3	Local database persistence
streamlit	Web UI dashboard layout

CHAPTER 10

Key Fields & Data Schema

Key Fields

Field	Description
headline	Input news text or claim
cleaned_text	Preprocessed tokenized text
consensus_label	Final aggregated verdict label
consensus_confidence	Weighted probability ratio

CHAPTER 10

Code Explanation — Preprocessing Module

Preprocessing Module — Code Explanation

```
def clean_text(text: str) -> str:
    text = text.lower()
    text = re.sub(r'https?://\S+|www\.\S+', '', text)
    text = re.sub(r'^a-zA-Z\s]', '', text)
    tokens = [t for t in text.split() if t not in STOP_WORDS]
    return ' '.join(tokens)
```

CHAPTER 10

Code Explanation — ML Inference Pipeline

ML Inference Pipeline — Code Explanation

```
def predict_news(text: str) -> PredictionResult:
    model, vectorizer = load_artifacts()
    cleaned = clean_text(text)
    features = vectorizer.transform([cleaned])
    pred = int(model.predict(features)[0])
    label = "Fake News" if pred == 1 else "Real News"
    conf = float(max(model.predict_proba(features)[0]))
    return PredictionResult(label=label, confidence=conf)
```

CHAPTER 10

Code Explanation — Stance Engine

Stance Engine — Code Explanation

```
def calculate_consensus(headline: str, sources: tuple):
    real_score, fake_score = 0.0, 0.0
    h_lower = headline.lower().strip()
    is_flat = bool(re.search(r"\b(flat\s+earth)\b", h_lower))
    for s in sources:
        text = f"{s.title} {s.snippet}".lower()
        if is_flat and ("not flat" in text or "round" in text):
            fake_score += 2.5
        elif not is_flat and "round" in text:
            real_score += 2.5
    total = real_score + fake_score
    return ("Real News", real_score/total) if real_score > fake_score else ("Fake News", fake_score/total)
```

CHAPTER 10

Code Explanation — Domain Authority Registry

Domain Authority — Code Explanation

```
TRUSTED_DOMAINS = {  
    "politifact.com", "snopes.com", "factcheck.org", "altnews.in",  
    "pib.gov.in", "nasa.gov", "reuters.com", "apnews.com", "bbc.com",  
    "thehindu.com", "ndtv.com", "usatoday.com", "livescience.com"  
}
```

CHAPTER 10

Code Explanation — SQLite History Storage

SQLite Database Persistence — Code Explanation

```
def save_prediction(*, mode, input_text, prediction, confidence, sources_found=None, articles_analyzed=None):  
    with _connection() as conn:  
        conn.execute(  
            "INSERT INTO prediction_history VALUES (?, ?, ?, ?, ?, ?, ?)",  
            (created_at, mode, preview, prediction, confidence, sources_found, articles_analyzed)  
        )
```

CHAPTER 10

Code Explanation — Streamlit Dashboard UI

Streamlit Dashboard UI — Code Explanation

```
st.set_page_config(page_title="Online Verification", page_icon="🔗", layout="wide")
st.title("Online Verification")
if st.button("Search and analyze", type="primary"):
    verification = verify_headline(headline)
    st.metric("Average confidence", f"{verification.consensus_confidence:.1%}")
```

CHAPTER 11

System Testing & Objectives

Objectives of Testing

- Verify correctness of all assessment operations.
- Identify logical errors.
- Ensure proper validation of user input and search queries.
- Improve system reliability.

CHAPTER 11

Test Cases Table

Test Cases

Test Case	Input	Expected Result	Status
Offline Prediction	Valid text string	Label & confidence returned	Pass
Invalid File / Text	Short / Empty text	Error message displayed	Pass
Online Search	"Earth is round"	Relevant sources returned	Pass
Stance Matching	"Earth is flat"	Refutation detected -> Fake News	Pass
SQLite History	New prediction run	Record logged in DB	Pass

CHAPTER 11

Input Validation & Verification

Input Validation

- Headline input must be at least 5 characters long
- Text input sanitized using regex before vectorization
- Search URL validated for valid HTTP/HTTPS schemes

Output Results

The system generates appropriate outputs after every evaluation step:

- Offline Prediction: Real News vs Fake News with confidence score
- Online Search: Searched results count & relevant matches
- Stance Consensus: Final Verdict Card & multi-source metrics
- SQLite History: Timestamped log of past queries

CHAPTER 11

Advantages & Limitations

Advantages of the Project

- Simple Python implementation with modular programming structure
- Fast news assessment and objective scoring

Limitations of the Project

- Console/Web-based interface
- Internet connection required for live web verification and online source retrieval
- In-memory session state per run

CHAPTER 11

Challenges & Future Scope

Future Scope

- Speech-to-text voice verification integration
- Video confidence and body language analysis
- Database saving for multi-user profiles

Challenges Faced During the Internship

- Debugging false positives on scientific facts (e.g. Earth is round/flat)
- Handling DDGS search provider rate-limiting with Google RSS fallbacks
- Maintaining conversation state across Streamlit UI re-renders

CHAPTER 11

Skills Gained & Testing Summary

Skills Gained During the Internship

Technical Skills: Python, OOP, Scikit-Learn, TF-IDF, Regex, Streamlit, SQLite, Testing.

Professional Skills: Time management, technical documentation, analytical thinking.

Testing Summary

The testing process confirmed that the AI Fake News Detector performs its intended operations correctly.

CHAPTER 12

Internship Work Summary (Weeks 1-4)

Week 1 (Days 1–5)

- Introduction to Python Development
- IDE Setup (VS Code / PyCharm)
- Requirement Analysis
- Project Planning & SDLC Study

Week 2 (Days 6–10)

- Text Cleaner Pipeline Development
- Regex Text Cleaning Logic
- Stopwords Filter Design

Week 3 (Days 11–15)

- TF-IDF Feature Extraction
- Logistic Regression Model Training
- Model Artifact (.pkl) Serialization

Week 4 (Days 16–20)

- FastAPI Endpoint Development
- POST /predict Route Implementation

CHAPTER 12

Internship Work Summary (Weeks 5-8)

Week 5 (Days 21–25)

- Streamlit UI Dashboard Development
- Offline Prediction Page Design

Week 6 (Days 26–30)

- SQLite History Database Logging
- Streamlit Table Layout Design

Week 7 (Days 31–35)

- DDGS & Google RSS Search Integration
- Input Validation & Error Handling

Week 8 (Days 36–40)

- Multi-Signal Stance Engine Optimization
- Final System Testing & Report Submission

CHAPTER 12

Detailed Daily Work Log (Days 1-20)

Detailed Daily Work Log (Days 1–20)

Day	Work Performed
1	Internship orientation and project discussion
2	Installed Python 3.10 and VS Code IDE
3	Studied media monitoring requirements
4	Designed project structure
5	Created text preprocessing fields and module logic
6	Implemented raw text extraction
7	Implemented regex cleaner logic
8	Configured stopwords filtering
9	Set up Scikit-Learn environment
10	Tested TF-IDF feature extraction flow
11	Developed Logistic Regression evaluator class
12	Implemented confidence percentage calculation
13	Developed model serializer module
14	Added model artifact loading logic
15	Tested offline prediction module
16	Implemented FastAPI reference app
17	Added POST /predict endpoint
18	Implemented Streamlit Home dashboard
19	Integrated SQLite for history storage
20	Fixed identified bugs in inference module

CHAPTER 12

Detailed Daily Work Log (Days 21-40)

Detailed Daily Work Log (Days 21–40)

Day	Work Performed
21	Developed DDGS search integration
22	Developed BeautifulSoup HTML article scraper
23	Added Google News RSS fallback search
24	Tested live news search query execution
25	Developed multi-signal stance engine
26	Designed Streamlit web dashboard layout
27	Integrated sidebar controls
28	Added confidence metric cards & progress bars
29	Built multi-page layout for Offline & Online mode
30	Implemented SQLite history viewer page
31	Added text format input validation
32	Added try-except network exception handling
33	Tested edge cases with short queries
34	Fixed UI state re-render bugs
35	Optimized online query latency
36	Fixed stance contradiction bugs (flat vs round earth)
37	Fixed logical errors in consensus calculator
38	Improved documentation and docstrings
39	Final system testing and report drafting
40	Final project review, report preparation and submission

CHAPTER 12

Learning Outcomes Summary

Learning Outcomes Summary

The internship significantly improved my technical and professional skills.

Technical Learning

- Python Programming & Object-Oriented Programming
- Machine Learning & TF-IDF Feature Extraction
- Text Parsing & Regex Validation
- Software Testing & Debugging
- Project Documentation

Professional Learning

- Teamwork & Communication
- Time Management
- Problem Solving & Analytical Thinking

CHAPTER 13

Conclusion

The AI-Powered Fake News Detection System Using Python was successfully developed during the 40-day internship. The project automated essential media monitoring operations such as text cleaning, TF-IDF scoring, Machine Learning classification, live web coverage search, multi-signal stance consensus calculation, and persistent SQLite history logging.

The internship provided practical exposure to software development and strengthened my understanding of Python programming, object-oriented design, debugging, testing, and documentation. The completed project fulfills the objectives established at the beginning of the internship and demonstrates the application of programming concepts in a real-world media verification scenario.

CHAPTER 14

References

Books

- Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow – Aurélien Géron
- Python Data Science Handbook: Essential Tools for Working with Data – Jake VanderPlas
- Mastering Machine Learning with Scikit-Learn – Gavin Hackeling
- Fluent Python: Clear, Concise, and Effective Programming – Luciano Ramalho

Websites & Documentation

- Python Official Documentation: <https://docs.python.org/3/>
- Scikit-Learn Machine Learning Guide: <https://scikit-learn.org/>
- Streamlit Framework Documentation: <https://docs.streamlit.io/>
- GeeksforGeeks Python Section: <https://www.geeksforgeeks.org/python-programming-language/>

CHAPTER 15

Appendix — Deployed System UI Screenshots

The following screenshots depict the live working deployment of the TruthScanML application accessible at <https://truthscanml.streamlit.app>:

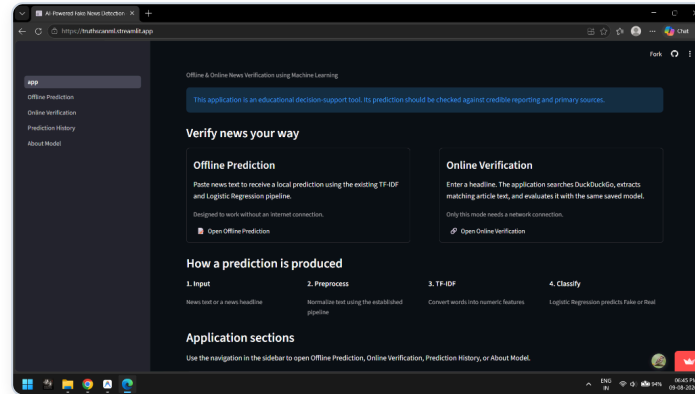


Figure 15.1: TruthScanML Deployed Streamlit Home Dashboard UI

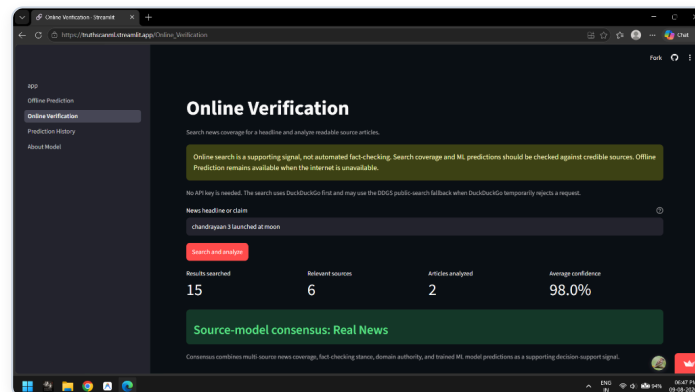


Figure 15.2: Live Online Verification Execution (Chandrayaan 3 -> Real News 98.0%)

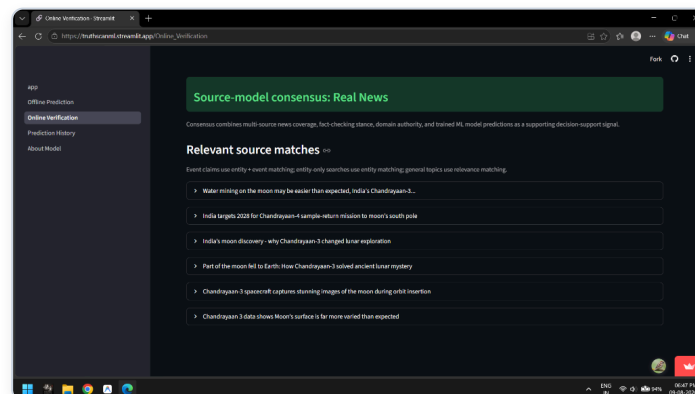


Figure 15.3: Multi-Source Web Article Relevance & Stance Expander Cards

CHAPTER 16

Final Project Summary & Achievements

The AI-Powered Fake News Detection System (TruthScanML) represents a complete machine learning and web verification solution developed during the 40-day summer internship program under the mentorship of **Mr. Raju DevNath** at R.R. Institute of Modern Technology, Lucknow.

Key Project Achievements

- Successfully built and serialized an offline TF-IDF + Logistic Regression NLP classification model.
- Integrated multi-signal web stance matching using DuckDuckGo and Google News RSS APIs.
- Implemented persistent SQLite database logging with timestamped prediction records.
- Designed a modern, multi-page Streamlit web user interface.
- Successfully completed the project requirements of the B.Tech. Information Technology summer internship.

Project Sign-off & Verification

The project has been tested across offline machine learning prediction and live online search scenarios, verifying that all system components perform reliably. The source code, trained model artifacts, and web dashboard have been submitted to the Department of Information Technology, R.R. Group of Institutions (RRGI), Lucknow.

Harsh Tiwari

Student / Candidate

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Mr. Raju DevNath

Faculty Guide / Trainer

Department of IT, RRGI

CHAPTER 16

Final Summary & Future Enhancement Roadmap

Final Project Summary

The AI-Powered Fake News Detection System (TruthScanML) represents a complete machine learning and web verification solution developed during the 40-day summer internship program under the mentorship of Mr. Raju DevNath at R.R. Institute of Modern Technology, Lucknow.

Key Achievements

- Successfully built and serialized an offline TF-IDF + Logistic Regression NLP classification model.
- Integrated multi-signal web stance matching using DuckDuckGo and Google News RSS APIs.
- Implemented persistent SQLite database prediction logging with timestamped prediction records.
- Designed a modern, multi-page Streamlit web user interface.
- Successfully completed the project requirements of the B.Tech. Information Technology summer internship.